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Improving Recognition of Overlapping Human Speech on Single-Microphone Social Robots with Streaming Two-Mask CMGAN Post-Filtering

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Author Contributions

Yue Li: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Visualization, Writing – Original Draft, Project Administration.

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All authors reviewed and approved the final manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Ethics Statement

The research ethics self-check of the Ethics Review Committee of the Faculty of Science (BETHCIE), Vrije Universiteit Amsterdam, completed on 01/02/2025, indicated that this project does not require further evaluation by the Research Ethics Review Committee. All participants provided informed consent prior to recording.

Data and Code Availability

Code, scripts, and anonymized derived materials supporting this study will be made available in a public GitHub repository. Raw participant audio recordings are not publicly released because they may contain identifiable voice data; access may be considered upon reasonable request subject to consent, ethics, and privacy constraints.